

Decimal multiplication (decimal $\times$ integer + decimal $\times$ decimal)

Unit 4 · decimal operation (multiplication) · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Unit 4 + Modern Education Volume 5A Unit 13

Core traps: ☐ T2 Decimal point positioning - summing of digits · $0.3 \times 0.4 = 0.12 \neq 1.2$ · Multiply 10/100/1000 shift

SSPA related: ● High frequency The sub-test must be taken every year, accounting for about 12-15% of Paper 1

Prerequisite knowledge: P4 addition and subtraction of decimals, P4 decimals

Our goals: ① Decimal $\times$ whole number (digit method) ② Decimal $\times$ decimal (digit method) ③ $\times 10/100/1000$ vs $\times 0.1/0.01$ ④ Decimal multiplication problem

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Calculate $24 \times 3 = ?$ Then calculate $2.4 \times 3 = ?$ (Tip: first treat as $24 \times 3 = 72$, then deal with the decimal point)	Basic	
2	Calculate $1.25 \times 4 = ?$ (Hint: First assume $125 \times 4 = 500$, 1.25 has 2 decimal places $\rightarrow$ there are 2 answers $\rightarrow 5.00 = 5$)	Basic	
3	Number of decimal places: 0.3 How many decimal places are there? What about 0.25? What about 0.125? (Key: Addition of digits for multiplication)	Basic	
4	Convert 0.3×0.4 to $3 \times 4 = 12$. How many decimal places do the two factors have? What should the answer be?	Basic	
5	Calculate $3.2 \times 10 = ?$ $3.2 \times 100 = ?$ $3.2 \div 10 = ?$ (Observe the moving direction and number of digits of the decimal point)	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Decimal $\times$ integer - "Count as an integer first, then count the decimal places" ● SSPA

Decimal $\times$ integer three-step method:

- ① Calculate as an integer first |||SEP|||: Ignore the decimal point and calculate as an integer multiplication
Count the decimal places |||SEP|||: How many decimal places does the original decimal have?
- ② Point the decimal point |||SEP|||: Count the same number of digits from right to left in the product, point the decimal point |||SEP|||: How many decimal places does the original decimal have?
- ③ decimal point: Count the same number of digits from right to left in the product, and put the decimal point
example: $2.4 \times 3 =$ Calculate first $24 \times 3 = 72$, 2.4 has 1 decimal place $\rightarrow 72 \rightarrow 7.2$

❏ Trap detonation example

Calculation: $2.4 \times 3 = ?$

✗ Common mistakes

$2.4 \times 3 = 72$ (missing decimal point!)

Forget to count decimal places and hit the decimal point

✓ Correct solution

$2.4 \times 3 = 7.2$

$24 \times 3 = 72$, 2.4 has 1 decimal place $\rightarrow 7.2$

🧠 Tip: "Cover the decimal point and count as an integer. After counting the digits, count back from the right."

⚠ The most common mistake: forgetting to add the decimal point after multiplying! Remember: Decimal $\times$ Integer = Calculate as an integer first, then click the decimal point at the end.

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	Calculate $2.4 \times 3 = ?$ ($24 \times 3 = 72$, decimal places: 1 $\rightarrow$ answer = 7.2)		
Example 2	Calculate $1.25 \times 4 = ?$ ($125 \times 4 = 500$, decimal places: 2 $\rightarrow$ answer = 5.00 = 5)		

Knowledge point 2: Decimals

Method for summing decimals $\times$ decimal digits:

① **Calculate as integers first:** Ignore all decimal points, calculate as integers $\times$ integers

② **Sum of digits:** Count the number of decimal places for each of the two factors, **Add** to get the total number of digits

③ **|||SEP|||** : Count the total number of digits in the product from right to left, with the decimal pointPad zero

④ : If there are not enough digits in the product, pad zero in frontExample:
 $0.3 \times 0.4 \rightarrow 3 \times 4 = 12$, decimal places $(1+1) = 2 \rightarrow 0.12$  $0.3 \times 0.4 = 0.12$, not 1.2!
(This is the trap that most people make mistakes!)(This is the trap most people make wrong!)

 **Trap comparison questions (core of this class)**

Calculation: $0.3 \times 0.4 = ?$

 **Typical mistakes (70% of students make mistakes the first time)**

$0.3 \times 0.4 = 1.2$

35%+ students intuitively believe that $0.3 \times 0.4 = 1.2$
(multiplied by two decimals should become larger? Wrong!)

 **Correct solution (digit summing method)**

$0.3 \times 0.4 = 0.12$

$3 \times 4 = 12$, each of the two factors has 1 decimal place, a total of 2 $\rightarrow$ 2 digits from the right $\rightarrow 0.12$

 **Tip: "Decimals $\times$ decimals, the digits must be added up; $0.3 \times 0.4 = 0.12$, not 1.2!"**

 **Fatal Trap:** Multiplying two decimals less than 1 will result in a result smaller than both factors! $0.3 \times 0.4 = 0.12$
($0.12 < 0.3$ and $0.12 < 0.4$)

 **Zero padding trap:** $0.2 \times 0.3 = 0.06$ ($3 \times 2 = 6$, total decimal places = 2, you need to pad a zero in front of 6 $\rightarrow 0.06$)

Knowledge Point 2 Worked Examples

#	Question	Difficulty	Working Space
Example 3	Calculate $0.3 \times 0.4 = ?$ (Total number of digits: $1+1=2$, $3 \times 4 = 12 \rightarrow 0.12$)		

Example
4

Calculate $0.25 \times 0.3 = ?$ (Total number of digits: $2+1=3$, $25 \times 3=75 \rightarrow 0.075$)



Knowledge point 3: Multiply 10/100/1000 vs multiply 0.1/0.01/0.001 ● SSPA Advanced

Decimal point shifting rule: |||SEP|||: The decimal point moves 1 place to the

① $\times 10$ right(the number becomes larger) |||SEP|||: The decimal point moves 2 places to the right |||SEP|||: The decimal point moves to the

② $\times 100$ rightShift 3 places : The decimal point is shifted 1 place to

③ $\times 1000$ left|||SEP||| LeftShift 3 bits (= $\div 1000$)

④ $\times 0.1$: Decimal point directionLeftShift 1 position (= $\div 10$, the number becomes smaller)

⑤ $\times 0.01$: Decimal point directionLeftShift 2 bits (= $\div 100$)

⑥ $\times 0.001$: Decimal point directionLeftShift 3 bits (= $\div 1000$)



$\times 10/100/1000$

Move the decimal point to the right



$\times 0.1/0.01/0.001$

Decimal point moved to the left



Not enough digits

pad zero in front



Check calculation

$\times 10$ becomes larger $\times 0.1$ becomes smaller

example

Example 5: Calculate $3.2 \times 10 = ?$ $3.2 \times 100 = ?$ $3.2 \times 1000 = ?$ (Move the decimal point to the right)

example

Example 6: Calculate $56 \times 0.1 = ?$ $5.6 \times 0.01 = ?$ $56 \times 0.01 = ?$ (Move the decimal point to the left)

Knowledge point 4: Multiplication of decimals word problems ● SSPA required test

Multiplication of decimals word problems - pay attention to the units and decimal places:

① "Per unit A $\times$ quantity B" = total quantity (for example: \$12.5 per kilogram $\times$ 3.2 kilograms)

② Area of the rectangle = length $\times$ width (multiply two decimals)

③ The number of decimal places in the answer depends on the question requirements (usually keep 1-2 digits or write the exact value)

example

Example 7: A bag of rice weighs 2.5 kilograms. How many kilograms does the total weight of 3 bags weigh?

example

Example 8: A book is 0.8 centimeters thick. How many centimeters is the 0.5 book thick?

 **Formula (shift the decimal point): "Multiply by ten to the right, multiply by zero and one to the left; the number of digits is equal to the number of zeros."**

III. Lesson layered synchronization practice (total 15 questions)

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
9	$1.2 \times 3 = ?$		
10	$0.5 \times 4 = ?$		
11	$2.3 \times 2 = ?$		
12	$0.2 \times 0.3 = ?$		
13	$0.4 \times 0.5 = ?$		

Advanced layer (total 5 questions,   choose do)

#	Question	Difficulty	Working Space
14	$2.5 \times 0.4 = ?$		
15	$1.25 \times 0.8 = ?$		
16	$0.15 \times 0.3 = ?$ (Note: The sum of the digits = $2+1=3$, do you need to add zeros?)		
17	$4.2 \times 10 = ?$ $4.2 \times 100 = ?$ $4.2 \times 1000 = ?$		
18	$56 \times 0.1 = ?$ $56 \times 0.01 = ?$ $5.6 \times 0.01 = ?$		

 challenge layer (total 5 questions,   choose do)

#	Question	Difficulty	Working Space
19	$0.25 \times 0.16 = ?$		
20	$0.12 \times 0.03 = ?$ (Total number of digits = $2+2=4$, $12 \times 3=36$, how many zeros need to be added?)		
21	$3.6 \times 2.5 = ?$		
22	Comparing 0.3×0.5 and 0.03×5 , which one is larger? Why? (Caution: Different sums of digits result in different decimal point positions!)		
23	It is known that $24 \times 35 = 840$, without recalculating, directly write $0.24 \times 0.35 = ?$		

IV. decimal multiplication application question (total 6 question , all must-do)

#	Question	Difficulty	Working Space
24	A pack of sugar weighs 1.2 kilograms. If you buy 0.5 packs, how many kilograms does it weigh in total?		
25	Each meter of fabric sells for \$15.8. How much does it cost to buy 2.5 meters?		
26	A rectangle is 2.4 meters long and 1.5 meters wide. What is the area in square meters?		
27	13.6 kilometers per liter of petrol, how many kilometers per liter of petrol?		
28	A pen costs \$3.50. How much does it cost to buy 8 pens? (Decimal $\times$ Integer)		
29	My brother ran 4.5 laps. Each lap was 0.4 kilometers long. How many kilometers did he run in total?		

V. Ultimate challenge area (total 3 question)

#	Question	Difficulty	Working Space
 1	It is known that $125 \times 8 = 1000$, without recalculating, write the following results directly: (a) $1.25 \times 0.08 = ?$ (b) $0.125 \times 80 = ?$ (c) $12.5 \times 0.8 = ?$		
 2	The water tank has a capacity of 2.4 liters and is filled with 0.35 liters of water per minute. How many liters of water are there in the tank after 0.5 minutes of filling? (First find the amount of water injection, and add the decimal times the number of decimal digits!)		
 3	Rectangle A: length 2.4cm, width 0.5 times the length. Rectangle B: The length is 1.5 times of A and the width is 0.8 times of A. What is the area of B? (Three-step reasoning: first find the width of A $\rightarrow$ find the length of B $\rightarrow$ find the width of B $\rightarrow$ area)		

VI. Class after homework

Basic must-do questions (total 5 questions)

#	Question	Difficulty	Working Space
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H1	$0.6 \times 7 = ?$		
H2	$3.4 \times 2 = ?$		
H3	$0.5 \times 0.4 = ?$		
H4	$0.7 \times 0.3 = ?$		
H5	$2.5 \times 0.6 = ?$		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	A pencil is 18.5 cm long. How many centimeters are there in total if 6 pencils are connected end to end? (Decimal $\times$ Integer)		
H7	Calculate using the sum of digits method: (a) $0.25 \times 4 = ?$ (b) $2.5 \times 0.4 = ?$ (c) $25 \times 0.04 = ?$ Observe the three answers, what patterns did you find?		
H8	A rectangle is 3.5 meters long and 1.2 meters wide. What is the area in square meters?		

VII. The Lessoncorecommon errorssummary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Decimal point positioning error : $0.3 \times 0.4 = 1.2$	Sum of digits: $3 \times 4 = 12$, 1 digit + 1 digit = 2 digits $\rightarrow 0.12$ (not 1.2!)
2	When adding digits, count only one factor	Count the decimal places of both factors and then add them. $0.25 \times 0.3 \rightarrow 2+1=3$ digits.
3	Forgot to add zeros : $0.2 \times 0.3 = 0.6$	$2 \times 3 = 6$, 2 decimal places in total $\rightarrow$ 1 zero needs to be added $\rightarrow 0.06$ (not 0.6)
4	$\times 10$ moved to the right but in the wrong direction	$\times 10/100/1000 \rightarrow$ Decimal point Move right (the number becomes larger)
5	$\times 0.1$ Wrong direction SEP : Thought it was to the right: Thought it was to the right	$\times 0.1 = \div 10 \rightarrow$ Decimal point Move left (number becomes smaller)

6	The zero at the end of the integer was forgotten SEP : 1.25×4=5.00 not 500 :1.25×4=5.00 not 500	The zero at the end of the product, after the decimal point, can be omitted: 5.00 = 5
7	Missed application question unit conversion	For application questions, you must write "Answer:..." + the correct unit, and the amount must be kept to 2 decimal places.